

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Class Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Dave's Taxonomy: Articulation (Level 4)
Question Count: 10

Total Marks: 30

Code of Conduct

I Nandhini sri V(192521344) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks Class Test

1. Analyze a multinational retail organization struggling with fragmented customer data across online and offline channels, and propose a unified big data architecture that enables real-time customer analytics and personalized recommendations.
2. Evaluate the effectiveness of distributed computing models in processing large-scale scientific research data, and justify how parallel processing improves computational efficiency and scalability compared to centralized systems.
3. Design a fraud detection framework for a digital payment platform using big data analytics, and justify how streaming data processing and predictive models improve fraud prevention accuracy.
4. Assess the challenges of storing and processing multimedia data such as video, audio, and images in a big data environment, and recommend suitable technologies and storage mechanisms for efficient management.
5. Construct a disaster recovery and business continuity strategy for a cloud-based big data platform, and justify how redundancy, replication, and automated failover mechanisms ensure operational resilience.
6. Analyze the impact of data quality issues such as inconsistency, duplication, and missing values in a large-scale analytics platform, and propose data governance strategies to improve analytical reliability.
7. Design a scalable recommendation system for an online streaming service using big data technologies, and justify how user behavior analytics and distributed processing enhance recommendation accuracy and user engagement.
8. Evaluate the role of cloud computing in modern big data ecosystems, and critically examine its advantages and limitations in terms of scalability, cost management, flexibility, and security.

9. Develop a real-time analytics framework for transportation and logistics management, and justify how big data technologies improve route optimization, fuel efficiency, and operational decision-making.
10. Critically analyze ethical and legal challenges associated with big data collection and analytics, and recommend organizational policies that ensure responsible data usage, regulatory compliance, and user trust.

Class Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Accuracy	30%	Accurate and well-expressed technical answers	Mostly accurate answers	Partial accuracy	Inaccurate answers
Coverage	20%	Covers all required points	Covers most points	Limited coverage	Very poor coverage
Use of Examples	15%	Uses relevant and meaningful examples	Some relevant examples	Weak examples	No useful examples
Analytical Awareness	20%	Shows strong awareness of issues and implications	Reasonable awareness	Basic awareness	Minimal awareness
Clarity	15%	Clear and concise writing	Generally clear	Some unclear expressions	Poor clarity

1. Analyze a multinational retail organization struggling with fragmented customer data across online and offline channels, and propose a unified big data architecture that enables real-time customer analytics and personalized recommendations.

Unified Big Data Architecture for Retail

A multinational retail company gets customer data from websites, mobile apps, shops, billing systems and social media. Since the data is available in different places, the company may find it difficult to understand customers.

A unified big data architecture can solve this problem. First, data is collected from all sources. Then it is stored in a common storage system such as a data lake. Big data tools like Apache Spark can process the data. Real-time processing can be used to understand customer activities immediately.

The company can analyze previous purchases, searches and interests to recommend suitable products.

Example: If a customer searches for a mobile phone, the system can recommend a mobile phone case, charger or similar phone.

Thus, a unified architecture helps combine customer data, understand customer behavior, provide personalized recommendations and increase sales.

2. Evaluate the effectiveness of distributed computing models in processing large-scale scientific research data, and justify how parallel processing improves computational efficiency and scalability compared to centralized systems.

Distributed Computing in Scientific Research:

- Scientific research produces huge amounts of data. Processing all this data using a single computer takes more time. Distributed computing can solve this problem.
- In distributed computing, a large task is divided into smaller tasks and given to multiple computers. These computers process the tasks at the same time. This is called parallel processing.

- For example, weather research collects data from satellites and weather stations. Different computers can process different parts of the data simultaneously.
- Distributed computing provides faster processing, better scalability and efficient use of resources. More computers can be added when the data increases.
- Therefore, distributed computing is more suitable than a centralized system for large-scale scientific research.

3. Design a fraud detection framework for a digital payment platform using big data analytics, and justify how streaming data processing and predictive models improve fraud prevention accuracy.

Fraud Detection Framework for a Digital Payment Platform:

Digital payment platforms handle a large number of transactions every day. Some of these transactions may be fraudulent. Big data analytics can help the payment platform identify suspicious transactions quickly.

The fraud detection process can be:

Transaction Data → Data Collection → Real-Time Processing → Fraud Analysis → Alert/Block

First, the system collects information such as transaction amount, location, time, device details and previous transaction history.

Next, streaming data processing checks transactions in real time. This helps the system identify unusual activities immediately.

Predictive models can also be used to study previous fraudulent transactions. These models can identify similar patterns in new transactions and calculate the possibility of fraud.

Example: If a customer normally makes small payments in Chennai but suddenly makes a very large payment from another location, the system can identify the transaction as suspicious.

If a transaction is considered risky, the system can send an alert, ask for additional verification or temporarily block the transaction.

The main benefits are:

- ❖ Quick fraud detection
- ❖ Reduced financial loss
- ❖ Improved transaction security

- ❖ Better detection accuracy
- ❖ Real-time monitoring
- ❖ Increased customer trust

4. Assess the challenges of storing and processing multimedia data such as video, audio, and images in a big data environment, and recommend suitable technologies and storage mechanisms for efficient management.

Challenges of Managing Multimedia Data in Big Data

Multimedia data includes **images, videos, audio files and other media files**. These types of data are usually very large and are difficult to store and process in a big data environment.

Main Challenges

1. **Large Data Size** – Videos, images and audio files require a large amount of storage space.
2. **Different Formats** – Multimedia files are available in different formats such as JPG, PNG, MP4 and MP3.
3. **Processing Time** – Processing large videos and images requires more computing power and time.
4. **Fast Data Growth** – New multimedia data is continuously created and uploaded.
5. **Data Transfer** – Large files require more network bandwidth and take more time to transfer.

Suitable Technologies

Cloud storage can be used to store large multimedia files. **Hadoop Distributed File System (HDFS)** can divide the data and store it across multiple computers. **Apache Spark** can be used to process large amounts of data quickly.

Example

A video streaming company stores millions of videos. Cloud storage can provide enough space to store these videos, while distributed processing can be used to analyze video data and user activities.

Benefits

- Provides large storage capacity
- Handles increasing data
- Improves processing speed
- Supports scalability

- Makes data management easier

Conclusion

Multimedia data creates storage and processing challenges because of its large size and different formats. Using **cloud storage, HDFS and Apache Spark** can help organizations store and process multimedia data efficiently.

5. Construct a disaster recovery and business continuity strategy for a cloud-based big data platform, and justify how redundancy, replication, and automated failover mechanisms ensure operational resilience.

Disaster Recovery and Business Continuity Strategy

A cloud-based big data platform stores important business data. Problems such as **server failure, cyberattacks, network failure or natural disasters** can stop the system. Therefore, a disaster recovery and business continuity strategy is needed.

Main Strategies

1. **Backup** – Important data should be backed up regularly. If the original data is lost, it can be restored from the backup.
2. **Replication** – Copies of important data should be stored in different locations. If one location fails, another copy can be used.
3. **Redundancy** – Extra servers and storage systems should be available so that the system can continue working when one component fails.
4. **Automatic Failover** – If the main server fails, the system can automatically switch to a backup server.
5. **Monitoring** – The system should be continuously monitored to identify failures quickly.

Example

For example, if the main server of an online shopping application fails, the backup server can automatically take over. Customers can continue using the application without a long interruption.

Benefits

- Reduces system downtime
- Prevents data loss
- Provides continuous service
- Improves system reliability
- Helps recover quickly

- Protects important business information

Conclusion

Backup, replication, redundancy and automatic failover are important for disaster recovery. They help a cloud-based big data platform continue its operations even when failures occur.

6. Analyze the impact of data quality issues such as inconsistency, duplication, and missing values in a large-scale analytics platform, and propose data governance strategies to improve analytical reliability.

Data Quality Issues and Data Governance

Big data analytics depends on good-quality data. If the data contains errors, duplicate records or missing values, the final analysis may not be accurate. This can lead to wrong business decisions.

Common Data Quality Issues

1. **Inconsistency** – The same information may be stored differently in different systems.
2. **Duplication** – The same customer or transaction may be stored more than once.
3. **Missing Values** – Some important information may be missing from the dataset.
4. **Incorrect Data** – Some records may contain wrong or outdated information.

Impact on Analytics

Poor-quality data can cause:

- Wrong analysis
- Incorrect predictions
- Poor business decisions
- Loss of money
- Reduced customer satisfaction

Data Governance Strategies

The organization can improve data quality by using:

1. **Data Cleaning** – Remove duplicate and incorrect data.
2. **Data Validation** – Check the data before storing it.
3. **Data Standards** – Use the same format and rules for storing data.
4. **Regular Monitoring** – Check the data regularly for errors.
5. **Data Security** – Protect data from unauthorized access.

Example

Suppose the same customer is stored three times in a database. The company may think that there are three different customers. Data cleaning can identify and remove the duplicate records.

Conclusion

Good-quality data is necessary for reliable big data analytics. Data cleaning, validation, standardization and regular monitoring can improve data quality and help organizations make better decisions.

7. Design a scalable recommendation system for an online streaming service using big data technologies, and justify how user behavior analytics and distributed processing enhance recommendation accuracy and user engagement.

Scalable Recommendation System for an Online Streaming Service

An online streaming service has millions of users who watch different types of movies, videos or songs. A recommendation system can study user behavior and suggest content based on their interests.

The system can collect data such as **watch history, search history, ratings, likes, dislikes and viewing time**. This data is stored and processed using big data technologies.

The process can be:

User Activity → Data Collection → Data Storage → Data Processing → Recommendation

Big data tools can process information from a large number of users. Distributed processing helps the system handle increasing users and large amounts of data.

The system can study what each user watches and recommend similar content.

Example: If a user watches many action movies, the system can recommend other action movies or newly released action content.

The recommendation system provides the following benefits:

- Gives personalized content
- Improves recommendation accuracy
- Saves users' search time
- Increases viewing time
- Improves user satisfaction
- Increases user engagement

Therefore, big data technologies and user behavior analysis help streaming services provide better recommendations and handle millions of users efficiently.

8. Evaluate the role of cloud computing in modern big data ecosystems, and critically examine its advantages and limitations in terms of scalability, cost management, flexibility, and security.

Role of Cloud Computing in Modern Big Data Ecosystems:

Cloud computing plays an important role in big data because organizations need a large amount of storage and computing power to process big data.

Cloud computing provides resources such as **storage, servers and processing power through the internet**. It helps organizations store and process large amounts of data without purchasing many physical servers.

Advantages

1. **Scalability** – Cloud resources can be increased when the amount of data increases.
2. **Cost Management** – Organizations can pay for the resources they use instead of buying expensive hardware.
3. **Flexibility** – Cloud services can be accessed from different locations and devices.
4. **Large Storage** – Cloud platforms provide large storage capacity for big data.
5. **Easy Management** – The cloud provider manages many hardware and infrastructure requirements.

Limitations

1. **Security** – Data stored in the cloud may face security risks if it is not properly protected.
2. **Privacy** – Sensitive customer data needs proper privacy protection.
3. **Internet Dependency** – Cloud services require a reliable internet connection.
4. **Cost Increase** – Poor resource management can result in higher cloud costs.

Example

For example, an online shopping company may have normal traffic on regular days but very high traffic during festival sales. Using cloud computing, the company can increase its resources during the sale and reduce them later.

Conclusion

Cloud computing provides **scalability, flexibility, storage and cost benefits** for big data applications.

However, organizations must properly manage **security, privacy and cost** to use cloud computing effectively.

9. Develop a real-time analytics framework for transportation and logistics management, and justify how big data technologies improve route optimization, fuel efficiency, and operational decision-making.

Real-Time Analytics Framework for Transportation and Logistics

Transportation and logistics companies handle many vehicles, deliveries and routes every day. Big data analytics can help these companies use real-time information to improve their operations.

The framework can be:

Data Collection → Data Processing → Real-Time Analysis → Decision Making

Data can be collected from **GPS devices, vehicles, traffic systems, weather information and delivery applications**.

The collected data is processed in real time to understand the current location of vehicles, traffic conditions and possible delays.

Based on this information, the system can suggest the best route for the vehicle. This helps reduce travel distance and time. Fuel consumption can also be monitored to identify unnecessary fuel usage.

Example: If a delivery vehicle is travelling on a road with heavy traffic, the system can detect the traffic and suggest an alternative route.

Benefits

- Reduces travel time
- Saves fuel
- Improves delivery speed
- Reduces traffic delays
- Improves vehicle utilization
- Helps in quick decision-making
- Reduces operational costs

Therefore, real-time big data analytics helps transportation and logistics companies improve **route optimization, fuel efficiency and overall operational decision-making**.

10. Critically analyze ethical and legal challenges associated with big data collection and analytics, and recommend organizational policies that ensure responsible data usage, regulatory compliance, and user trust.

Ethical and Legal Challenges of Big Data

Big data allows organizations to collect and analyze a large amount of information about users. It is useful for business and decision-making, but it also creates ethical and legal problems.

Main Challenges

1. **Privacy** – Personal information may be collected and used without proper understanding by users.
2. **Data Security** – Personal data can be stolen or accessed by unauthorized people.
3. **Lack of Consent** – Users should know what data is being collected and why it is being used.
4. **Data Misuse** – Organizations should not use customer data for purposes that were not communicated to users.
5. **Legal Issues** – Organizations must follow applicable data protection and privacy laws.
6. **Bias** – Incorrect or biased data can produce unfair results.

Recommended Policies

Organizations should follow proper data privacy and security policies. They should collect only the required data and obtain user consent where necessary.

Data should be protected using **encryption, passwords and access control**. Regular security checks should also be conducted.

Employees should be trained about data privacy and responsible data usage. The organization should also regularly check whether its data practices follow legal requirements.

Example

For example, an online shopping website may collect customer information to provide personalized product recommendations. The company should inform customers about the use of their data and protect it from unauthorized access.

Conclusion

Big data should be used responsibly. **Privacy protection, user consent, data security, legal compliance and proper data policies** help organizations use big data safely and maintain user trust.